MA 403 Real Analysis
Homework 13

Consider the set $\mathcal{C}([0,2\pi])$ of all continuous functions $f : [0,2\pi] \to \mathbb{R}$.

1. Prove that $\mathcal{C}([0,2\pi])$ is a vector space over $\mathbb{R}$.

2. Consider the function $\|\cdot\|_{\sup} : \mathcal{C}([0,2\pi]) \to \mathbb{R}$, given by

$$\|f\|_{\sup} := \sup\{f(x) \,:\, x \in [0,2\pi]\}.$$

Prove that $\|\cdot\|_{\sup}$ defines a norm on the vector space $\mathcal{C}([0,2\pi])$.

3. Consider the metric $d_{\sup}$ on $\mathcal{C}([0,2\pi])$ defined by

$$d_{\sup}(f,g) := \|f-g\|_{\sup}.$$

If $f,g \in \mathcal{C}([0,2\pi])$ are defined by $f(x) = \sin(x)$, $g(x) = \cos(x)$, find the values of:

(a) $\|f\|_{\sup}$ (b) $\|g\|_{\sup}$ (c) $d_{\sup}(f,g)$.

4. Let $\{f_n\}$ be a sequence of functions defined on an interval I. Let $x \in I$ be a point and suppose that each f_n is continuous at x. Suppose that $\{f_n\}$ converges uniformly to f on I. Show that if $\{x_n\}$ is a sequence of points in I satisfying $x_n \to x$ as $n \to \infty$, then $f_n(x_n) \to f(x)$ as $n \to \infty$.
Is the conclusion still true if the convergence $f_n \to f$ is not uniform?

5. In class we proved that if $\{f_n\}$ is a sequence of functions which are Riemann-integrable on a compact interval $[a,b]$, and $f_n \to f$ uniformly on $[a,b]$, then $\lim_{n\to\infty} \int_a^b f_n(t)dt = \int_a^b f(t)dt$.

Show that the theorem fails for improper integrals, by producing a sequence $\{f_n\}$ of continuous functions on $[0,\infty)$, such that each f_n vanishes outside a bounded interval, and such that $\int_a^b f_n(t)dt = 1$ for each n, but for which $\{f_n\}$ converges uniformly to 0 on $[0,\infty]$.

6. Consider the sequence of functions $\{f_n\}$, where $f_n(x) = \sqrt{x^2 + \frac{1}{n}}$.

(a) Identify the pointwise limit function f.

(b) Prove that the sequence $\{f_n\}$ converges uniformly to the limit function f.

(c) Is f differentiable? What goes wrong? Does $\{f_n'\}$ converge uniformly?